

JEE MAIN REVISION MASTER: HALOALKANES AND HALOARENES 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	SN1 and SN2 Nucleophilic Substitution	10 PYQs	Reaction Mechanisms
High	Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	9 PYQs	Elimination & Aromatic

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
SN1 and SN2 Nucleophilic Substitution	$\text{SN1: Rate} \propto [\text{RX}]$; $(3^\circ > 2^\circ > 1^\circ)$, Racemization , $\text{SN2: Rate} \propto [\text{RX}][\text{Nu}]$; $(1^\circ > 2^\circ > 3^\circ)$, Inversion
Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	Saytzeff Rule: Highly substituted, more stable alkene is the major product in dehydrohalogenation

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 SN1 and SN2 Nucleophilic Substitution (High Yield — 10 PYQs)

Core Principle: Kinetics, stereochemical outcomes, carbocation rearrangements in SN1, and concerted backside displacement in SN2.

■ **Caution / JEE Pitfall:** Polar PROTIC solvents (e.g. H_2O , ROH) stabilize carbocations and favor SN1. Polar APROTIC solvents (e.g. acetone, DMSO) favor SN2 by desolvating nucleophiles.

■ **Speed Trick:** Allylic and benzylic halides undergo BOTH SN1 (resonance-stabilized carbocation) and SN2 (low steric hindrance) at remarkably high rates.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "In $\text{S}_{\text{N}}(\text{N})1$ reaction, the alkyl halide that on hydrolysis produces racemic mixture is..." [Ans: B]
- [2025 Shift] Q#2 (MCQ): "The organometallic compound $(\text{CH}_3)_3\text{CMgBr}$ on reaction with D_2O produces..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "..." [Ans: N/A]

3.2 Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions (High Yield — 9 PYQs)

Core Principle: β -elimination mechanism, Saytzeff vs Hofmann regioselectivity, and unreactivity of haloarenes toward nucleophilic substitution.

■ **Caution / JEE Pitfall:** Haloarenes (like chlorobenzene) are extremely UNREACTIVE toward nucleophilic substitution due to partial double bond character (C-Cl resonance) and sp^2 carbon.

■ **Speed Trick:** Presence of electron-withdrawing groups (like $-NO_2$) at ortho- and para-positions dramatically increases reactivity of haloarenes toward nucleophilic substitution.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is..." [Ans: D]
- [2026 Shift] Q#3 (MCQ): "The compound from which chlorobenzene cannot be prepared easily is..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Chlorobenzene reacts with bromine gas in the presence of Anhyd $AlBr_3$ to yield p-Bromochlorobenzene. This reaction is classified as _____" [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is

- (A) Longer and stronger
- (B) Shorter and stronger
- (C) Shorter and weaker
- (D) Longer and weaker [Correct]

Correct Answer: D

Solution Analysis: The C-Cl bond in methyl chloride is compared with the C-Cl bond in chlorobenzene. In methyl chloride, CH_3Cl , carbon is sp^3 hybridised and the C-Cl bond is a normal single bond. In chlorobenzene, C_6H_5Cl , the chlorine atom is attached to an sp^2 hybridised carbon of the benzene ring. Also, due to resonance, the lone pair of chlorine overlaps with the benzene ring. This gives the C-Cl bond some partial double bond character. Because of this partial double bond character, the

Benchmark Problem 2: KCET 2026 (Q#2)

In S_N1 reaction, the alkyl halide that on hydrolysis produces racemic mixture is

- (A) Tertiary butyl bromide
- (B) 2-bromobutane [Correct]
- (C) Isopropyl bromide
- (D) Methyl bromide

Correct Answer: B

Solution Analysis: The correct option is B. 2-bromobutane. In an S_N1 reaction, the alkyl halide first forms a carbocation. This carbocation is planar, so the nucleophile can attack from either side with almost equal chance. For 2-bromobutane, $CH_3-CHBr-CH_2-CH_3$ the carbon bearing bromine is a chiral carbon. During hydrolysis, bromide ion leaves and forms a planar carbocation: $CH_3-\text{vec}(C)H-CH_2-CH_3$. Water can attack this carbocation from both sides, producing both enantiomers of butan-2-ol: $\text{CH}_3-CH(OH)-CH_2-CH_3$.